BC local government finance model

Every municipality in British Columbia, 2019-2024: benchmarks, infrastructure renewal, a five-year financial plan, property tax by class and borrowing room. Now showing: Fernie.

Independent analysis of the Province of British Columbia's Local Government Statistics. Not affiliated with or endorsed by any municipality or by the Province.

How to use it

1. Pick a municipality in the yellow cell on the Dashboard. Every sheet follows.
2. Change the plan drivers on Plan: inflation, wages, how much of the renewal gap to close, and how much of it to borrow.
3. Test a levy change on Tax, and a borrowing amount on Debt.
4. Rank a municipality's tier on any metric on Peers.
5. Checks proves the numbers; Province shows how far the asset data can be trusted.

Sheets

Dashboard	Eight tiles against the peer tier, five sentences written by formulas, two charts
Profile	Six years of revenue, expenses, position, debt, assets and tax
Peers	Thirteen metrics with tier and province percentiles; the tier ranked on any of them
Assets	Schedule 503 by category; renewal need and the reinvestment ratio
Plan	A five-year financial plan and a what-if data table over its two big choices
Tax	Assessment, rates and multiples by class; a levy-to-rates calculator
Debt	The legal servicing limit, the room left, and what it would borrow
Province	Totals by year and the asset data-quality screen
Checks	17 controls, each with a tolerance

Sources

Province of British Columbia, Ministry of Municipal Affairs: Local Government Statistics, schedules 201-707, 2019-2024. Open Government Licence – British Columbia.
Pipeline: pipeline/ (bronze and silver, held to contracts), dbt/ (gold marts and 31 data tests). data/sources.json records every file's URL and SHA-256.

BC local government finance: Fernie

Municipality Fernie

City in the EKO regional district · population 6,000 · peer tier 2,500 to 9,999 residents (47 municipalities) · statements through 2024

Own-source share of revenue	Share of the legal debt-servicing limit used	Reserves against a year of cash spending	Share of asset life used up
89.3%	**15.2%**	**0.76x**	**41.2%**
tier median 77.3%	tier median 15.4%	tier median 1.83x	tier median 42.6%
better than the tier median	better than the tier median	weaker than the tier median	better than the tier median
percentile in tier: 98%	percentile in tier: 49%	percentile in tier: 2%	percentile in tier: 40%

Capital reinvestment against renewal need	Taxes and charges on a representative house	Municipal property tax per resident	Long-term debt per resident
0.96x	**$6,264**	**$2,326**	**$428**
tier median 1.06x	tier median $4,507	tier median $1,246	tier median $950
weaker than the tier median	weaker than the tier median	above the tier median	better than the tier median
percentile in tier: 44%	percentile in tier: 87%	percentile in tier: 96%	percentile in tier: 27%

What the numbers say

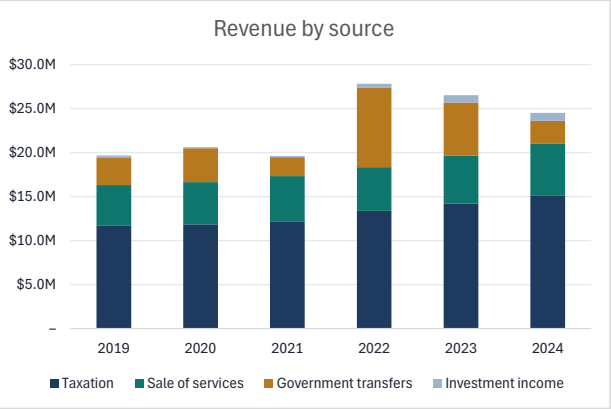
Fernie, 2024: revenue $24.5M, expenses $22.0M, a surplus of $2.5M (10.1% of revenue); 89% of revenue is its own.

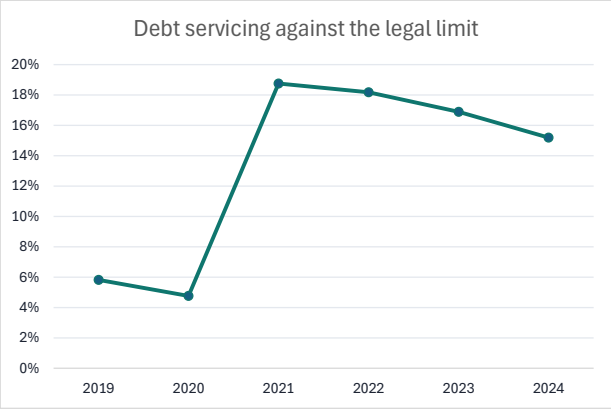
Fernie uses 15% of its legal servicing limit. The $4.8M a year still available would service about $62.9M of new borrowing at 4.3% over 20 years.

Capital additions cover 96% of the renewal need: $348K a year short of keeping pace with wear.

Closing 100% of the renewal gap by year 5, borrowing 30% of the new capital, needs municipal tax increases averaging 3.7% a year. The representative house's municipal tax goes from $2,842 to $3,406, and debt servicing reaches 14% of the legal limit.

A representative $907,932 house pays $6,264 in property taxes and charges, against a tier median of $4,507; 45% of it is the municipal share.





Independent analysis of the Province of British Columbia's Local Government Statistics (Open Government Licence – British Columbia). Not affiliated with or endorsed by any municipality or the Province. Built in Excel by Kush Patel.

Six-year profile: Fernie

Every line is the Province's schedule for that year, looked up for the selected municipality. A dash means the Province published nothing for that year.

Line	Column	2019	2020	2021	2022	2023	2024	Change, first to last
Revenue								
Taxation and grants in lieu	taxation	$11,679,941	$11,847,977	$12,126,305	$13,369,957	$14,194,174	$15,074,622	+29.1%
Sale of services	sale_of_services	$4,630,558	$4,796,932	$5,212,081	$4,985,343	$5,465,696	$5,910,458	+27.6%
Government transfers	government_transfers	$3,063,142	$3,861,857	$2,132,053	$9,031,310	$6,032,744	$2,633,114	-14.0%
Investment income	investment_income	$297,439	$99,811	$116,551	$448,469	$839,539	$897,000	+201.6%
Developer contributions	developer_contributions	$206,600	($144,741)	$80,000	–	–	–	-100.0%
Total revenue	**total_revenue**	**$19,992,680**	**$20,461,836**	**$19,666,990**	**$27,835,079**	**$26,532,153**	**$24,515,194**	**+22.6%**
Expenses								
Cash expenses	cash_expenses	$13,684,373	$12,791,061	$13,650,961	$16,539,897	$17,129,644	$17,764,865	+29.8%
Amortization	amortization	$3,256,048	$3,471,292	$3,476,738	$3,644,440	$4,162,923	$4,268,555	+31.1%
Total expenses	**total_expenses**	**$16,940,421**	**$16,262,353**	**$17,127,699**	**$20,184,337**	**$21,292,567**	**$22,033,420**	**+30.1%**
Annual surplus	annual_surplus	$3,052,259	$4,199,483	$2,539,291	$7,650,742	$5,239,586	$2,481,774	-18.7%
Financial position								
Financial assets	total_financial_assets	$10,154,160	$16,113,500	$20,138,489	$21,079,484	$28,409,540	$24,816,939	+144.4%
Liabilities	total_liabilities	$11,339,782	$13,813,880	$14,367,534	$15,411,959	$21,941,329	$16,880,536	+48.9%
Net financial assets	**net_financial_assets**	**($1,185,622)**	**$2,299,620**	**$5,770,955**	**$5,667,525**	**$6,468,211**	**$7,936,403**	**-769.4%**
Reserves and operating/capital surplus	financial_reserves	$1,525,280	$7,852,727	$11,185,508	$9,738,321	$10,859,555	$13,420,201	+779.9%
Long-term debt	long_term_debt	$2,686,653	$5,553,107	$4,814,554	$4,070,796	$3,321,143	$2,566,095	-4.5%
Debt capacity								
Revenue for the liability limit	revenue_for_limit	$16,741,329	$16,801,219	$17,578,972	$18,739,359	$20,973,687	$22,410,266	+33.9%
Liability servicing limit (25%)	liability_servicing_limit	$4,185,332	$4,200,305	$4,394,743	$4,684,840	$5,243,422	$5,602,567	+33.9%
Actual debt servicing	debt_servicing_cost	$243,269	$200,042	$824,208	$851,682	$885,711	$851,154	+249.9%
Share of the limit used	limit_used	5.8%	4.8%	18.8%	18.2%	16.9%	15.2%	+0.1 pts
Assets and tax								
Capital additions	tca_additions	$13,873,447	$4,209,779	$2,544,694	$10,814,019	$7,572,883	$5,282,136	-61.9%
Replacement value	replacement_value	$287,234,599	$296,225,505	$299,497,019	$313,485,667	$322,621,947	$365,516,873	+27.3%
Municipal property tax	municipal_property_tax	$10,743,925	$10,997,472	$11,502,422	$12,425,297	$13,139,811	$13,957,961	+29.9%
Representative house value	house_value	$527,027	$569,308	$619,538	$736,170	$876,753	$907,932	+72.3%
Taxes and charges, representative house	house_total_charges	$4,761	$4,954	$5,203	$5,491	$5,937	$6,264	+31.6%
Population (BC Stats estimate)	population	5,744	5,668	5,739	5,870	6,000	6,000	+4.5%

Peers: Fernie against 2,500 to 9,999 residents

Every active municipality in the same population tier, from the latest year. Percentiles count only municipalities that report the metric; 100% is the highest value, which is the best only where the direction says so.

The selected municipality on every metric

Metric	Key	Value	Tier median	Tier percentile	Province percentile	Direction	Reading
Revenue per resident	revenue_per_capita	$4,086	$3,328	67%	63%	context	above the tier median
Spending per resident	expenses_per_capita	$3,672	$2,784	89%	77%	context	above the tier median
Municipal property tax per resident	municipal_tax_per_capita	$2,326	$1,246	96%	92%	context	above the tier median
Own-source share of revenue	own_source_share	89.3%	77.3%	98%	86%	higher	better than the tier median
Annual surplus as a share of revenue	surplus_ratio	10.1%	21.7%	18%	17%	higher	weaker than the tier median
Share of the legal debt-servicing limit used	limit_used	15.2%	15.4%	49%	63%	lower	better than the tier median
Long-term debt per resident	debt_per_capita	$428	$950	27%	44%	lower	better than the tier median
Reserves and surplus per resident	reserves_per_capita	$2,237	$4,023	9%	18%	higher	weaker than the tier median
Reserves against a year of cash spending	reserves_to_cash_expenses	0.76x	1.83x	2%	6%	higher	weaker than the tier median
Share of asset life used up	asset_consumption_ratio	41.2%	42.6%	40%	38%	lower	better than the tier median
Capital reinvestment against renewal need	reinvestment_ratio	0.96x	1.06x	44%	49%	higher	weaker than the tier median
Taxes and charges on a representative house	house_total_charges	$6,264	$4,507	87%	80%	lower	weaker than the tier median
Municipal tax per $1,000 of assessment	effective_municipal_rate	4.83	4.33	62%	63%	lower	weaker than the tier median

The tier, ranked

Rank the tier by: Share of the legal debt-servicing limit used (limit_used)

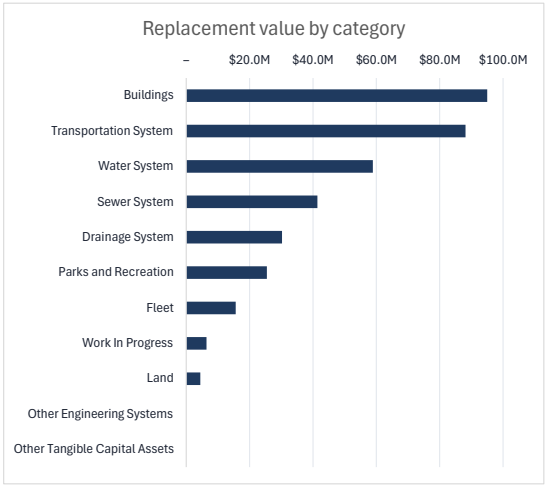
Municipality	Population	Value	Regional district
Golden	4,455	0.874	COL
Peachland	6,382	0.332	COK
Gibsons	5,098	0.294	SUN
Invermere	3,910	0.290	EKO
Trail	8,480	0.283	KOO
Oliver	5,824	0.278	OKS
Revelstoke	8,617	0.269	COL
Cumberland	4,901	0.264	COM
Bowen Island	4,181	0.261	MVRD
Merritt	7,809	0.250	TNR
Ladysmith	9,690	0.246	COW
Princeton	3,333	0.234	OKS
Elkford	2,805	0.231	EKO
Qualicum Beach	9,663	0.229	NAN
Rossland	4,390	0.222	KOO
Northern Rockies	4,400	0.193	
Pemberton	3,678	0.184	SQL
Tofino	2,653	0.182	ALB
Sicamous	3,005	0.177	COL
Osoyoos	5,730	0.162	OKS
Vanderhoof	4,839	0.161	BUL
Port McNeill	2,591	0.157	MW
Port Hardy	4,552	0.157	MW
Fernie	**6,000**	**0.152**	**EKO**
Castlegar	9,042	0.145	CKO
Armstrong	5,899	0.141	NOK
Chase	2,639	0.139	TNR
Sparwood	4,265	0.133	EKO
Anmore	2,533	0.126	MVRD
Creston	5,892	0.125	CKO
Kimberley	8,674	0.110	EKO
Enderby	3,323	0.108	NOK
Chetwynd	2,692	0.105	PEA
Lantzville	4,099	0.088	NAN
Highlands	2,698	0.076	CAP
Smithers	5,879	0.076	BUL
Hope	6,855	0.072	FVR
Grand Forks	4,447	0.066	KOO
Clearwater	2,751	0.046	TNR
Houston	3,268	0.042	BUL
Kent	7,201	0.026	FVR
Spallumcheen	5,858	0.014	NOK
Lake Cowichan	3,748	0.004	COW
Mackenzie	3,729	0.002	FFG
Duncan	5,513	0.000	COW
Metchosin	5,271	0.000	CAP

Infrastructure: Fernie, 2024

Schedule 503: what the municipality owns, what it would cost to replace, and whether capital spending keeps up with wear. The renewal need is re-derived, because the Province's own field is not usable as published (see Province).

Assets by category, latest year

Category	Historical cost	Accumulated amortization	Net book value	Additions	Replacement value	Life used up	Asset management plan	Replacement value share
Buildings	$27,847,711	$11,018,999	$16,828,712	$478,157	$94,976,440	40%	In Progress	26%
Transportation System	$46,235,209	$23,998,201	$22,237,008	$1,692,305	$88,140,126	52%	In Progress	24%
Water System	$38,989,296	$11,183,684	$27,805,612	$383,797	$58,886,174	29%	In Progress	16%
Sewer System	$28,934,056	$12,195,448	$16,738,608	$103,877	$41,426,092	42%	In Progress	11%
Drainage System	$12,295,932	$6,120,208	$6,175,724	$92,948	$30,242,160	50%	In Progress	8%
Parks and Recreation	$15,035,272	$1,922,661	$13,112,611	$2,548,597	$25,456,198	13%	In Progress	7%
Fleet	$14,478,454	$9,243,371	$5,235,083	$1,181,386	$15,594,339	64%	In Progress	4%
Work In Progress	$6,381,384	–	$6,381,384	($1,198,931)	$6,381,384	0%	Not applicable	2%
Land	$4,413,960	–	$4,413,960	–	$4,413,960	0%	Not applicable	1%
Other Engineering Systems	–	–	–	–	–	0%		0%
Other Tangible Capital Assets	–	–	–	–	–	0%	Not applicable	0%



Renewal: is capital spending keeping up with wear?

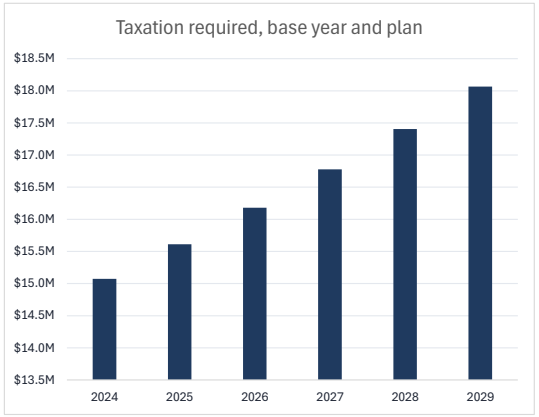
Data quality of the replacement values	plausible
Replacement value against historical cost	1.93x
Amortization this year	$4,268,555
Renewal need (amortization at replacement cost)	$8,237,308
Capital additions, three-year average	$7,889,679
Reinvestment ratio	0.96x
Annual renewal gap	$347,628
Share of replacement value under an asset management plan	0%
Share of asset life used up (depreciable assets)	41%

Capital additions cover 96% of the renewal need: $348K a year short of keeping pace with wear.

Five-year financial plan: Fernie

The plan the Community Charter (s.165) requires, built from the latest statements: costs grow with wages, prices and population; the renewal gap closes on a straight line to the target; new capital is funded from debt, reserves and taxation in the shares below. The output is the tax increase each year needs.

Drivers (change the yellow cells)			
Inflation (CPI)	2.5%	Base year (latest with statements)	2024
Wage and contract growth	3.5%	Population growth, two-year average	1.1%
Wage-like share of cash costs	55%	Cost growth (wages, prices, population)	4.2%
Share of the renewal gap closed by year 5	100%	Growth in revenue other than taxation	3.6%
Share of new capital borrowed	30%	Base capital program (three-year average additions	$7,889,679
Share of new capital from reserves	20%	Annual renewal gap at the base year	$347,628
MFA borrowing rate	4.30%		
Borrowing term (years)	20		



The plan

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Year	2024	2025	2026	2027	2028	2029
Cash operating expenses	$17,764,865	$18,503,096	$19,272,005	$20,072,866	$20,907,007	$21,775,812
Revenue other than taxation	$9,440,572	$9,780,958	$10,133,617	$10,498,992	$10,877,540	$11,269,737
Capital program	$7,889,679	$8,158,185	$8,435,185	$8,720,936	$9,015,703	$9,319,757
Capital above the base-year program	–	$268,506	$545,506	$831,257	$1,126,024	$1,430,078
New borrowing	–	$80,552	$163,652	$249,377	$337,807	$429,023
Servicing on new borrowing	–	$6,086	$18,449	$37,290	$62,811	$95,224
Drawn from reserves	–	$53,701	$109,101	$166,251	$225,205	$286,016
Capital funded from taxation	–	$134,253	$272,753	$415,628	$563,012	$715,039
Taxation required	**$15,074,622**	**$15,612,805**	**$16,179,919**	**$16,777,121**	**$17,405,619**	**$18,066,667**
Increase on the year before		+3.6%	+3.6%	+3.7%	+3.7%	+3.8%
Reserves and surplus, closing	$13,420,201	$13,366,500	$13,257,399	$13,091,147	$12,865,943	$12,579,927
Total debt servicing	$851,154	$857,240	$869,603	$888,444	$913,965	$946,378
Share of the legal limit used	15.2%	14.8%	14.5%	14.3%	14.1%	14.1%
Municipal tax on the representative house	$2,842	$2,943	$3,050	$3,163	$3,281	$3,406

Average tax increase a year, years 1 to 5 **+3.7%**

Closing 100% of the renewal gap by year 5, borrowing 30% of the new capital, needs municipal tax increases averaging 3.7% a year. The representative house's municipal tax goes from $2,842 to $3,406, and debt servicing reaches 14% of the legal limit.

What-if data table: average tax increase by gap closure and debt share

Gap closed ↓ Debt share →	+3.7%	0%	15%	30%	45%	60%
0%		+3.7%	+3.6%	+3.4%	+3.3%	+3.1%
25%		+3.8%	+3.6%	+3.5%	+3.3%	+3.2%
50%		+3.9%	+3.7%	+3.6%	+3.4%	+3.2%
75%		+4.0%	+3.8%	+3.6%	+3.4%	+3.3%
100%		+4.1%	+3.9%	+3.7%	+3.5%	+3.3%

Property tax by class: Fernie, 2024

Schedule 707. BC sets each class's rate as a multiple of the residential rate, so one number, the residential rate, sets them all. The calculator turns a levy change into the rates that raise it.

The classes

Property class	Assessment	Rate per $1,000	Class multiple	Variable-rate taxes	Flat and other taxes	Municipal taxes	Share of taxes	Assessment × multiple
Residential	$2,643,674,301	3.1307	1.00	$8,276,630	–	$8,276,630	58.8%	$2,643,674,301
Major Industry	–	0.0000	0.00	–	$3,461,569	$3,461,569	24.6%	–
Business/Other	$238,390,989	8.7573	2.80	$2,087,669	–	$2,087,669	14.8%	$666,831,237
Utilities	$3,182,720	33.8375	10.81	$107,695	$116,529	$224,224	1.6%	$34,399,458
Light Industry	$1,932,800	10.6445	3.40	$20,574	–	$20,574	0.1%	$6,571,525
Recreation	$1,207,000	3.1307	1.00	$3,779	–	$3,779	0.0%	$1,207,000
Farm	$14,438	3.1307	1.00	$45	–	$45	0.0%	$14,438
Managed Forest	–	0.0000	0.00	–	–	–	0.0%	–
Supportive Housing	$2	3.1307	1.00	–	–	–	0.0%	$2

Levy to rates

Variable-rate levy this year	$10,496,392
Assessment weighted by class multiples	$3,352,697,961
Residential rate the levy implies (per $1,000)	3.13073
Residential rate the Province publishes	3.13073
Levy change to test (defaults to plan year 1)	+3.6%
New residential rate	3.24250
Representative house value	$907,932
Municipal tax on it now	$2,842
Municipal tax on it after the change	$2,943

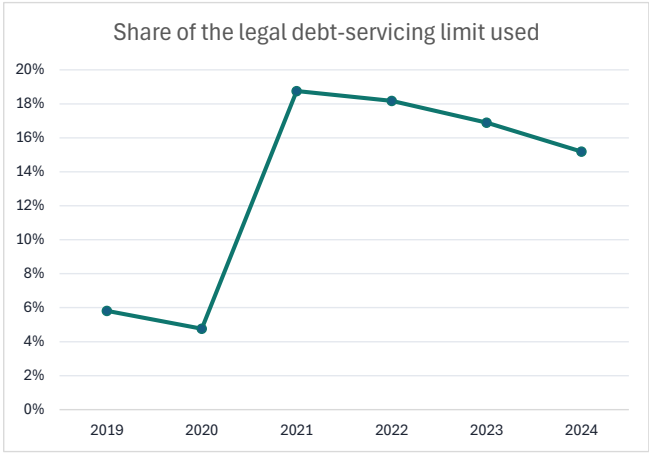
The levy and the class multiples imply a residential rate of 3.1307 per $1,000; the Province publishes 3.1307. A +3.6% levy change moves it to 3.2425, and the representative house's municipal tax from $2,842 to $2,943.

Borrowing room: Fernie, 2024

The Community Charter caps annual liability servicing at 25% of revenue (B.C. Reg. 254/2004). Schedule 602.1 reports the limit and what is used; the calculator turns the room left into a borrowing amount.

The limit	
Revenue for the purposes of the limit	$22,410,266
Liability servicing limit (25%)	$5,602,567
Actual debt servicing	$851,154
Servicing capacity available	$4,751,413
Share of the limit used	15.2%
Long-term debt outstanding	$2,566,095

Borrowing calculator	
Borrowing rate	4.30%
Term (years)	20
Amount to borrow	$10,000,000
Annual servicing on it	$755,496
Capacity left after it	$3,995,917
Most the remaining capacity could service	$62,891,268



Fernie uses 15% of its legal servicing limit. The $4.8M a year still available would service about $62.9M of new borrowing at 4.3% over 20 years.

Six years of servicing against the limit

Year	Limit	Servicing	Share used
2019	$4,185,332	$243,269	5.8%
2020	$4,200,305	$200,042	4.8%
2021	$4,394,743	$824,208	18.8%
2022	$4,684,840	$851,682	18.2%
2023	$5,243,422	$885,711	16.9%
2024	$5,602,567	$851,154	15.2%

The province: 161 municipalities, and how far their asset data can be trusted

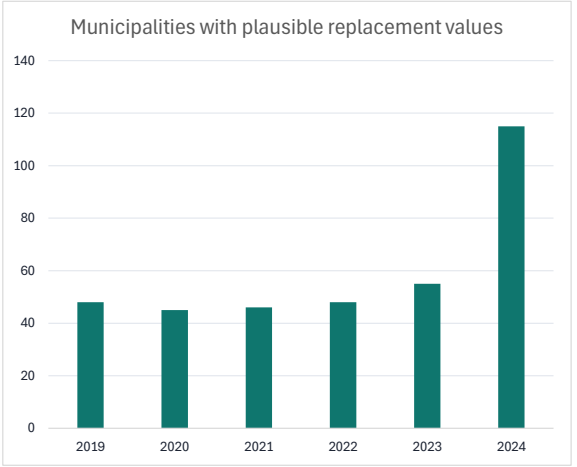
Totals and medians by year from mart_province_trend, and the data-quality screen behind every renewal figure in this workbook.

By year

Year	Municipalities	Revenue	Expenses	Long-term debt	Replacement value	Median limit used	Median house charges	Replacement values plausible
2019	162	$11,501.1M	$9,147.9M	$2,979.9M	$96,842.8M	11.9%	$3,557	48
2020	162	$11,508.4M	$9,024.3M	$2,981.8M	$98,413.9M	11.5%	$3,759	45
2021	161	$12,760.5M	$9,657.8M	$3,142.6M	$109,223.2M	10.8%	$3,952	46
2022	161	$13,889.8M	$10,451.0M	$3,353.5M	$109,933.4M	10.1%	$4,180	48
2023	161	$15,452.7M	$11,395.6M	$3,308.4M	$137,402.0M	10.1%	$4,458	55
2024	161	$16,286.4M	$12,313.3M	$3,557.7M	$191,113.8M	11.1%	$4,660	115

Latest year: the renewal picture

Municipalities whose replacement values pass the screen	131
...of which reinvest less than their renewal need	68
Their combined reinvestment ratio	1.07x
Municipalities reporting no replacement value	13
Municipalities with every asset class under a management plan	34
Municipality-years with no statements published (2019-2024)	9
Vancouver's reported renewal need, as a share of its replacement value	41%



In 2024, 131 municipalities report replacement values that pass the screen; 68 of them invested less over three years than their assets wore out. The field the Province labels annual replacement need is not usable as published: Vancouver's is 41% of its replacement value a year.

Checks: the model proves its own numbers

All 17 checks pass

#	Check	Expected	Actual	Tolerance	Status
1	Municipality-year rows loaded = dbt gold	968.	968.	0	PASS
2	Asset rows loaded = dbt gold	10,648.	10,648.	0	PASS
3	Tax-class rows loaded = dbt gold	8,712.	8,712.	0	PASS
4	Peer rows = active municipalities	161.	161.	0	PASS
5	Every active municipality is in the Dashboard list	161.	161.	0	PASS
6	Selection: financial assets less liabilities = net financial assets, every year	0.	0.	12	PASS
7	Selection: revenue less expenses = annual surplus, every year	0.	0.	12	PASS
8	Selection: the liability limit is 25% of revenue for the limit	5,602,566.5	5,602,567.	1	PASS
9	Levy and class multiples reproduce the published residential rate	3.13073	3.13072997	0.0001	PASS
10	Plan year 0 taxation = the base year's statements	15,074,622.	15,074,622.	0.01	PASS
11	Plan: taxation grows only by the cost, capital and revenue changes	0.	0.	0.01	PASS
12	Data table at the live drivers = the live plan	0.0368748	0.0368748	1E-12	PASS
13	Peer tier percentiles computed here = the pipeline's, all metrics	0.	0.	1E-09	PASS
14	Asset categories' replacement value = the municipality-year total	365,516,873.	365,516,873.	2	PASS
15	Power Query tiers cover every peer	161.	161.	0	PASS
16	Power Query categories cover the province's replacement value	191,113,796,372.	191,113,796,372.	1	PASS
17	Documented missing statements = municipality-years without revenue	9.	9.	0	PASS